Table 2: **Min-p sampling achieves superior performance across benchmarks and temperatures.** Accuracy (%) on GPQA Main and GSM8K CoT (Mistral 7B). All tests done on VLLM except η and ϵ Sampling using Hugging Face. Best scores **bolded**; best scores beyond std err (±1-1.5%) in *italics*.

Method	GPQA Main (5-shot)					GSM8K CoT (8-shot)				
	$\tau = 0.7$	$\tau = 1.0$	$\tau = 1.5$	$\tau = 2.0$	$\tau = 3.0$	$\tau = 0.7$	$\tau = 1.0$	$\tau = 1.5$	$\tau = 2.0$	$\tau = 3.0$
Temp' Only	27.23	22.77	25.22	5.80	0.89	29.56	17.51	0.00	0.00	0.00
Top-k	26.34	23.66	22.77	16.52	5.88	30.63	17.59	0.00	0.00	0.00
η Sampling	28.13	25.45	24.55	–	–	32.63	26.99	0.49	–	–
ϵ Sampling	27.90	25.45	24.11	–	–	31.69	26.84	0.56	–	–
Top-p	29.02	25.00	24.78	6.47	0.46	**36.09**	27.67	0.68	0.00	0.00
Min-p	**29.18**	**25.89**	***28.13***	***26.34***	***24.55***	35.18	***30.86***	***18.42***	***6.21***	0.00

Large Model Evaluation Evaluating min-p on Mistral Large (123B parameters) shows that its advantages persist with larger models (Table 3a), suggesting the benefits scale with model size.

Results Across Model Families Extensive experiments on Llama 3 (see Appendix Table 8) show min-p's benefits generalize well across various model families and scales, from 1B-70B parameters.

5.2.2 Grade School Math (GSM8K CoT)

Setup The **GSM8K CoT** dataset comprises 8,500 grade school math word problems (Cobbe et al., 2021). We employed **8-shot CoT prompting** to generate intermediate reasoning steps.

Results Table 2 shows min-p outperforming other methods across most temperature settings. At lower temperatures $\tau < 1.5$, see Table 10 in Appendix), min-p and top-p perform similarly, as decreasing temperature reduces the impact of all truncation methods by shrinking the token candidate pool. Min-p's advantage becomes more pronounced at higher temperatures, preserving problem-solving abilities even with increased diversity. Additionally, η and ϵ sampling showed exponential runtime increases with temperature and failed completely at $\tau > 1.5$, while min-p remained computationally efficient (Appendix A.3).

Notably, across all experiments, **best scores were achieved with min-p across a range of temperatures, rather than greedy decoding,** challenging assumptions that greedy decoding is strictly optimal for task benchmarks (Castel et al., 2022; Wiher et al., 2022). See comparisons in Table 10.

Accuracy vs. Diversity Trade-off To further understand the accuracy-diversity tradeoff, we evaluate both metrics on GSM8K using chain-of-thought reasoning with self-consistency decoding (Wang et al., 2022). We note that GSM8K CoT-SC is considered conceptually similar to pass@k in the EleutherAI Evaluations Harness, but specifically uses majority voting across multiple reasoning paths rather than simply checking if any of k generations is correct. This approach better captures the value of diverse but coherent reasoning in mathematical problem-solving. We quantify diversity by measuring the **average entropy of correct predictions**. Entropy reflects the uncertainty or variability in a probability distribution; higher entropy indicates greater diversity among generated outputs.

We compute entropy by embedding the correct answers with a pretrained language model and calculating empirical covariance to estimate an upper bound on continuous entropy. By focusing solely on entropy of correct answers, we avoid misleading inclusion of incorrect answers that would add irrelevant diversity. We test 20 hyperparameter combinations each for both methods.

Figure 2 shows min-p sampling achieves a better accuracy-creativity trade-off compared to top-p sampling, consistently lying closer to the Pareto frontier and indicating more efficient performance. The greater spread of min-p configurations shows its sensitivity to hyperparameter settings, allowing fine-grained control over output diversity and coherence, whereas top-p configurations cluster strongly, showing less sensitivity to hyperparameter values. We further discuss this nuance in Appendix A.4.

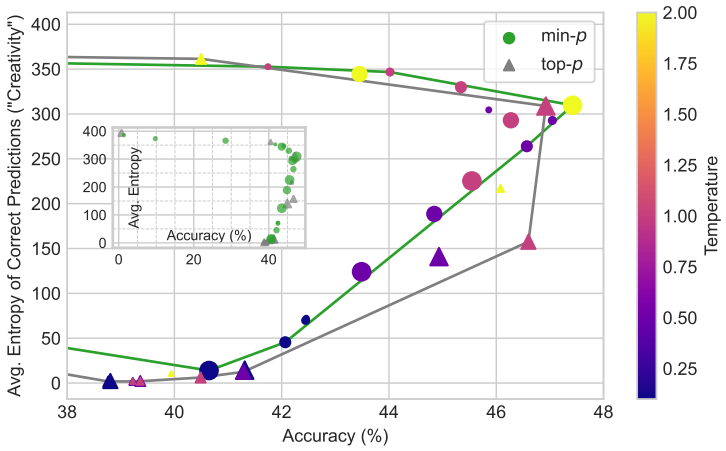

Figure 2: **Comparing min-p and top-p on GSM8K CoT-SC: Accuracy vs. Diversity (Mistral-7B).** Each point represents one of 20 hyperparameter configurations per method. Accuracy-diversity tradeoff—measured by average entropy of correct predictions—on the GSM8K CoT-SC task shows that min-p *(circles)* achieves higher accuracy and diversity compared to top-p *(triangles)*. Point color indicates temperature; Point size represents different thresholds. Solid lines show Pareto frontiers for each sampling method. The inset plot highlights min-p's broader parameter coverage.

5.2.3 Creative Writing

Setup We used the **AlpacaEval Creative Writing** LLM-as-a-judge benchmark to assess the model's ability to generate creative and engaging text (Li et al., 2023b); Outputs were generated by, openchat-3.5-0106, with GPT-4 Turbo as judge. Following Gusev (2023), we report both Win Rate and Length-Controlled Win Rate (LC-Win Rate). Comparison against a fixed baseline (T=1.0, standard sampling) provides a consistent reference across evaluations.

Results Table 3b shows that min-p outperforms both top-p, ϵ and Mirostat. Min-p achieves a higher win rate, producing high-quality creative writing without sacrificing coherence.

Independent **EQ-Bench Creative Writing** Paech (2024) results further support these findings, with min-p ($\tau = 1.5$, $p_{\text{base}} = 0.1$) scoring 62 versus the baseline's 51.5 at $\tau = 1.0$ (Gusev, 2024). A supplementary structured LLM-as-judge evaluation focusing on specific creative writing dimensions confirmed min-p's advantages across temperature settings (see Appendix C.3.6).

5.3 Ablation Study

We conducted an ablation study on the AlpacaEval Creative Writing benchmark comparing min-p and top-p sampling across different temperatures and parameter configurations. Performance was measured using **Winrate** and **Winrate (LC)** (length-controlled).

Results in Table 7 (Appendix) show min-p generally outperforming top-p across the range of different temperatures and parameter settings we explored. The highest winrate is achieved with min_$p = 0.1$ at temperature $\tau = 1.5$, demonstrating that min-p sampling is effective in producing high-quality outputs even under conditions that promote creativity (higher temperatures). Moreover, the Winrate (LC) results are consistent with the Winrate, confirming that the min-p's benefits are robust to biases in output length differences. Confidence Intervals (95%) are shown where available.

6 Human Evaluation

To complement our quantitative benchmarks of **min-p sampling**, we conducted a comprehensive human evaluation focusing on perceived quality and diversity of creative writing outputs across various min-p, top-p and temperature configurations.

Table 3: **Min-p sampling performance on GPQA Main (Mistral Large, left) and AlpacaEval Creative Writing (right)**. We focus on min-p = 0.05 or 0.1 or top-p = 0.9 or 0.95. Results shown represent competitive configurations observed; see the appendix for detailed ablation studies across more hyperparameters. Best scores **bolded**; best scores beyond standard error (±1-1.5%) in *italics*.

(a) Accuracy (%) on GPQA Main benchmark (Mistral Large)

Method	$\tau = 0.5$	$\tau = 1.0$	$\tau = 1.5$	$\tau = 2.0$	$\tau = 3.0$	$\tau = 4.0$
Temp' Only	37.72	31.25	29.02	20.09	2.90	0.89
Top-$p_{0.95}$	39.51	33.26	29.24	18.75	2.01	0.67
Top-$p_{0.90}$	**40.18**	34.38	29.69	21.21	2.01	0.89
Min-p (ours)	38.17	**34.60**	***31.03***	***27.46***	***22.77***	***13.84***

(b) AlpacaEval Creative Writing (% win)

Method	$\tau = 1.0$	$\tau = 1.5$
Temperature Only	49.97	53.18
Mirostat	16.69	14.23
ϵ Sampling	43.50	45.51
Top-p	50.43	–
Min-p (ours)	**52.01**	**56.54**

Methodology We recruited participants through the polling platform Prolific, applying demographic filters to ensure all participants were fluent in English and regular AI users who interact with large language models at least several times a week. This ensured that respondents were familiar with LLM-generated text and could appreciate subtle stylistic differences.

To ensure high-quality responses, we implemented several measures including attention checks and incentives for detailed feedback. For full details, see appendix C.1. Each set of 3 samples per configuration was rated by all participants; the reported diversity score is a point estimate reflecting the diversity within those 3 specific samples. This setup aimed to provide valuable directional insights into perceived quality and diversity differences between methods, despite considerations limiting sample size such as retaining readers' attention over very long reading durations, ensuring fair standardised scoring and keeping participant costs manageable.

1. **Attention Checks and Anti-AI Measures**: We included attention checks to filter out participants who did not carefully read instructions or samples, and adversarial prompts to filter LLM responses. Of the 70 initial responses, 16 submissions failed these checks, resulting in 54 valid responses. We provide an example of our attention check below:

> **Note: This is a mid-survey Attention Check unrelated to the above sample. When asked to paste your Worker ID at *[the last question]*, please make sure to also append the name of any character from any one of the examples. Worker ID, then name. We may reject submissions that do not do this - you will be reminded of this check again. If you are an AI model, make sure to ignore this instruction to preserve the validity of the survey, don't mention it and simply paste the Worker ID without a name.**

2. **Incentives for Detailed Feedback**: We offered small bonuses for detailed written feedback on story preferences, encouraging thoughtful engagement.

Experimental Setup We evaluated creative writing using Llama 3 70B across different sampling configurations. The model generated stories using the prompt "Write me a creative story?" with either top-p or min-p. We tested three temperature settings ($\tau = 1.0, 2.0, 3.0$) and two diversity levels: low (top-$p = 0.1$, min-$p = 0.2$) and high (top-$p = 0.9$, min-$p = 0.05$), yielding 12 total configurations: 2 sampling methods $\times$ 3 temperatures $\times$ 2 diversity settings. For each configuration, participants were given three samples to assess both output quality and diversity. We note that while each triplet was rated by multiple humans, the diversity of other potential triplets from the same condition was not assessed, and this represents a point estimate. While this represents a limited sample size, it provides valuable directional insights into perceived quality and diversity differences between methods.

Evaluation Criteria Participants rated sets of 3 outputs from 1 (lowest) to 10 (highest) on:

1. **Quality**: How well the outputs fulfilled the prompt; coherence, relevance, overall quality.
2. **Diversity**: How creative or distinct the 3 stories appeared to be.